

Maputnik

Making your own map with open source tools

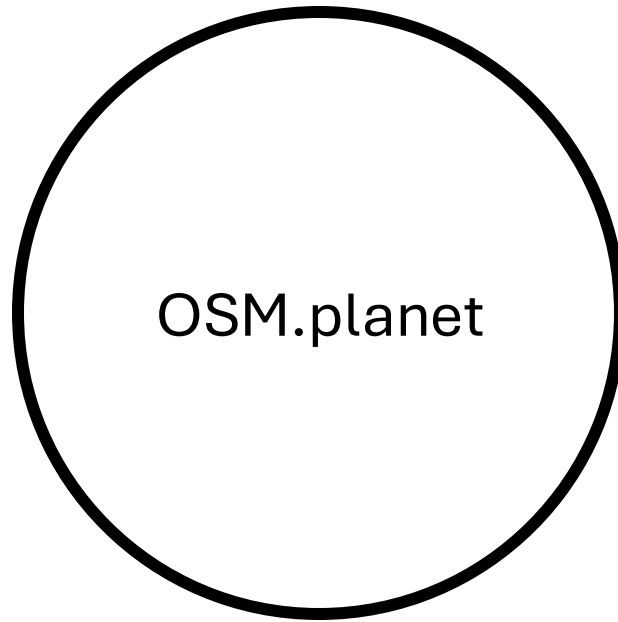
What is Maputnik?

“A free and open visual editor for the [MapLibre GL styles](#) targeted at developers and map designers.”

Put **simply** its an easy way to make your own maps with an **editor**
you can **easily** interact with in your browser

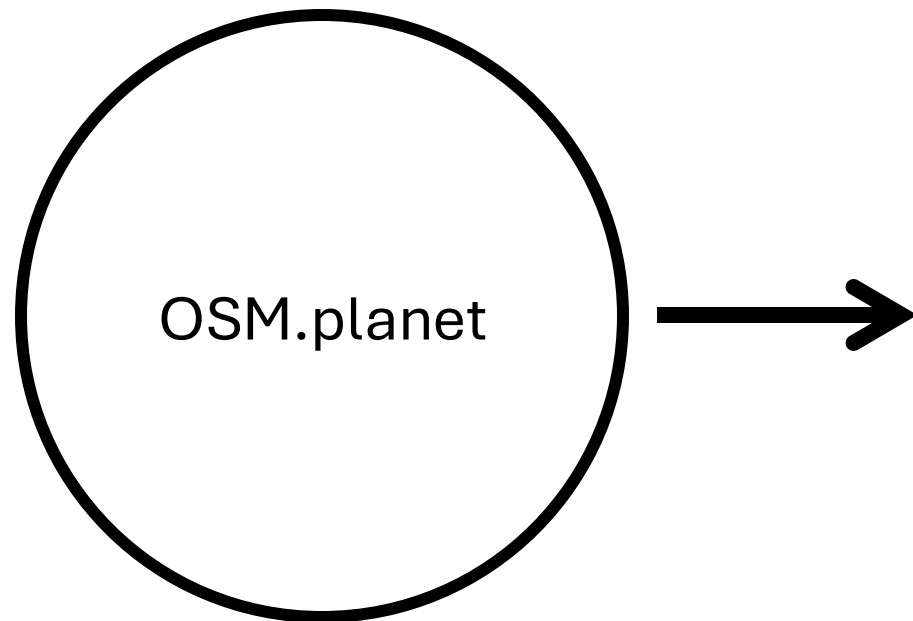
But what is **Maputnik**?

That's a little more complicated...

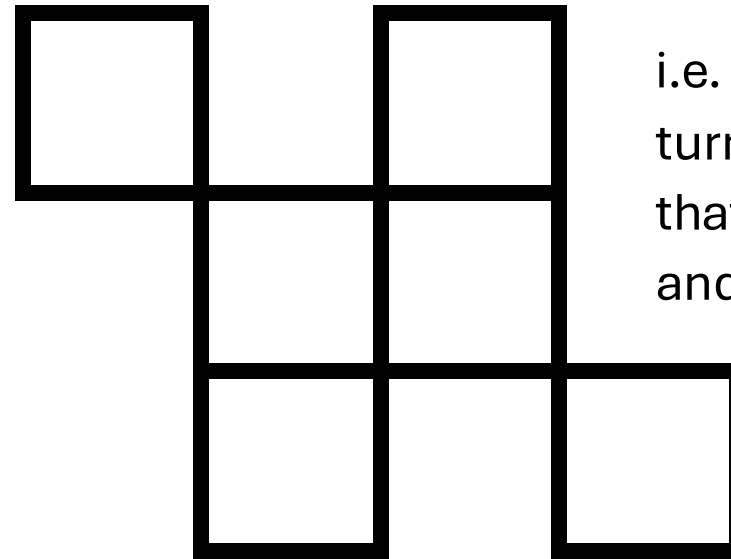


But what is **Maputnik**?

It used to be ...



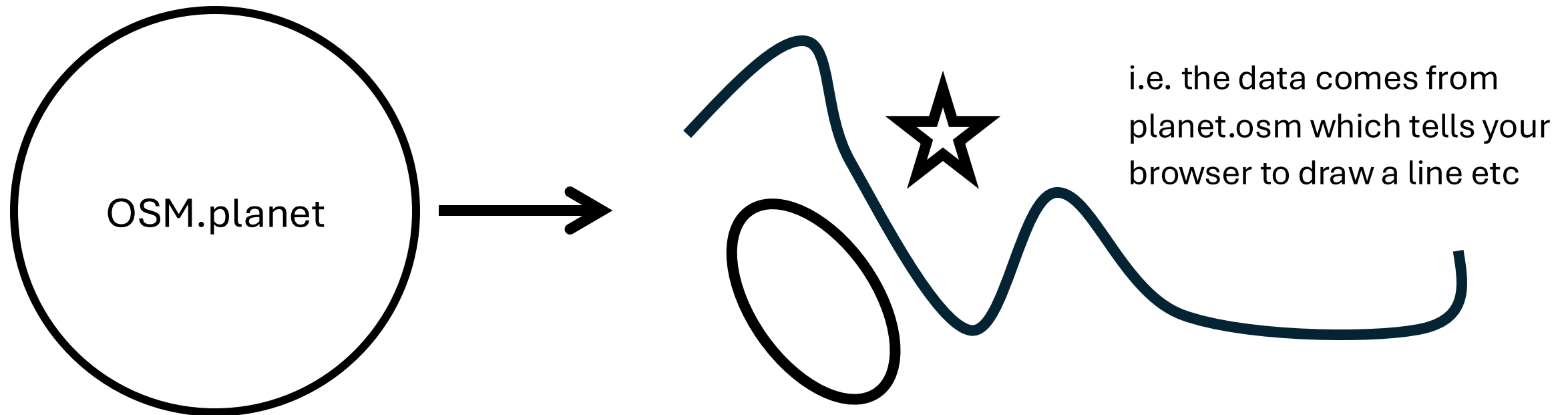
Pre-rendered map tiles



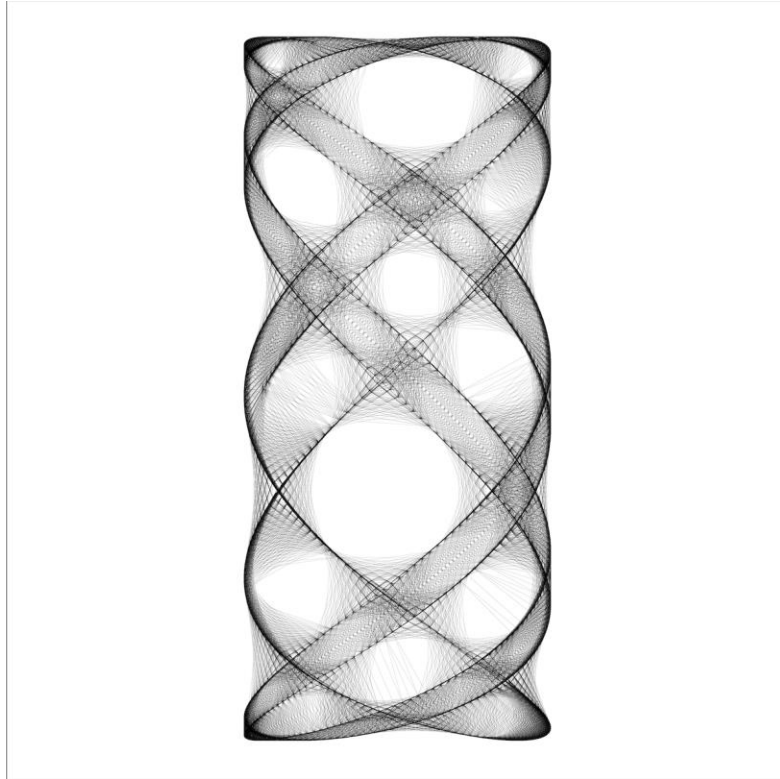
i.e. we would run code to
turn osm.planet into images
that you could then zoom in
and out of

But what is **Maputnik**?

Now ...



In principle this is the same as vectors in design software where you can keep zooming in



vector

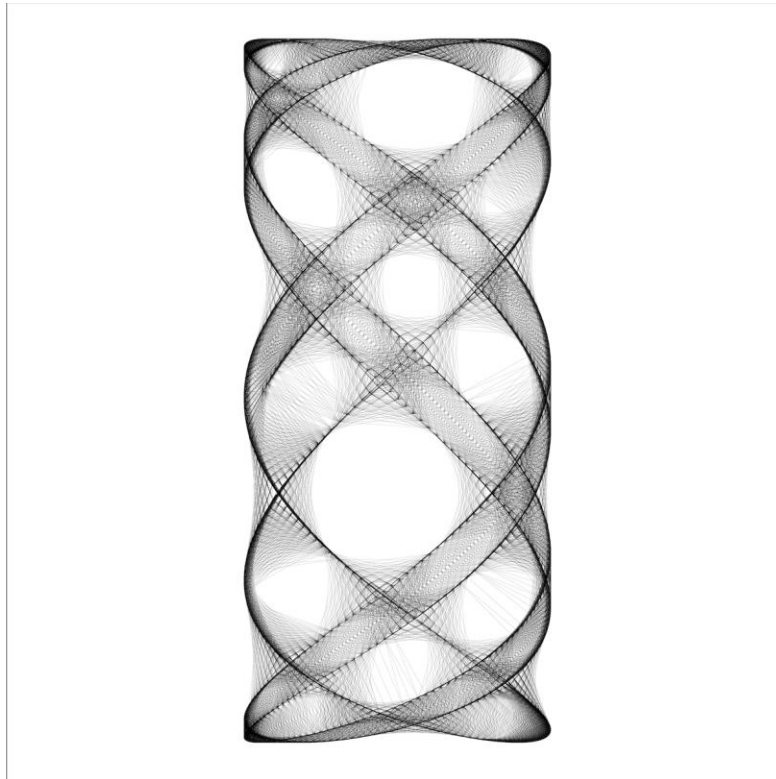
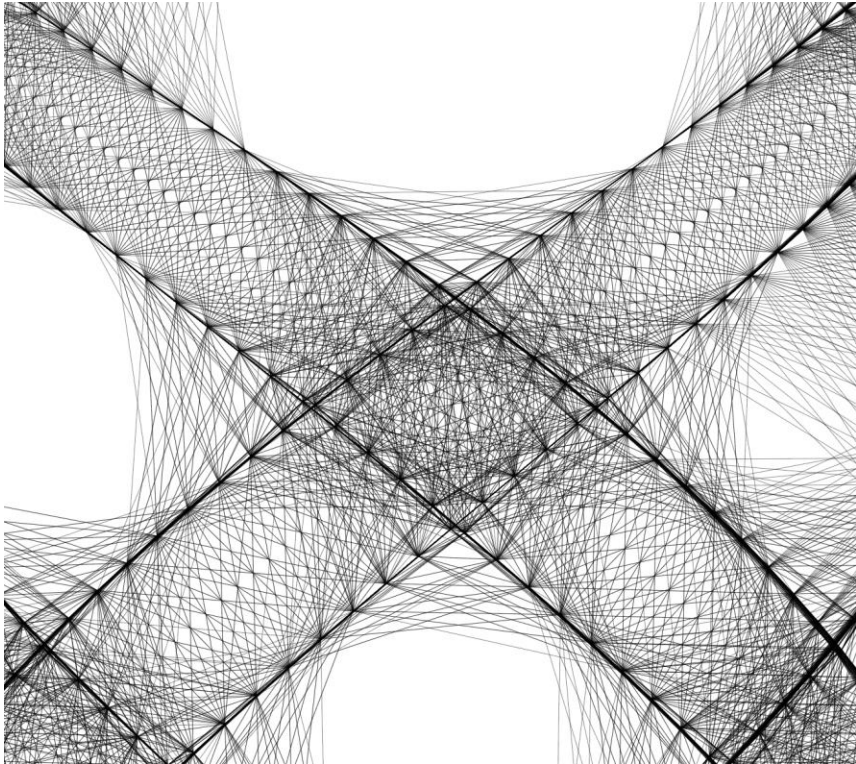


image file

In principle this is the same as vectors in design software where you can keep zooming in



vector

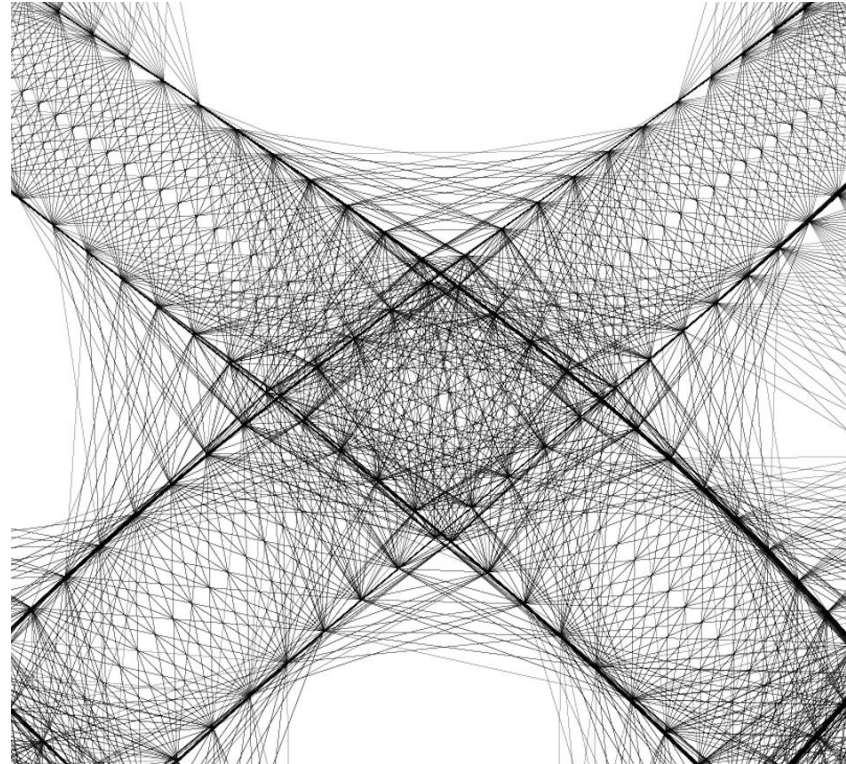
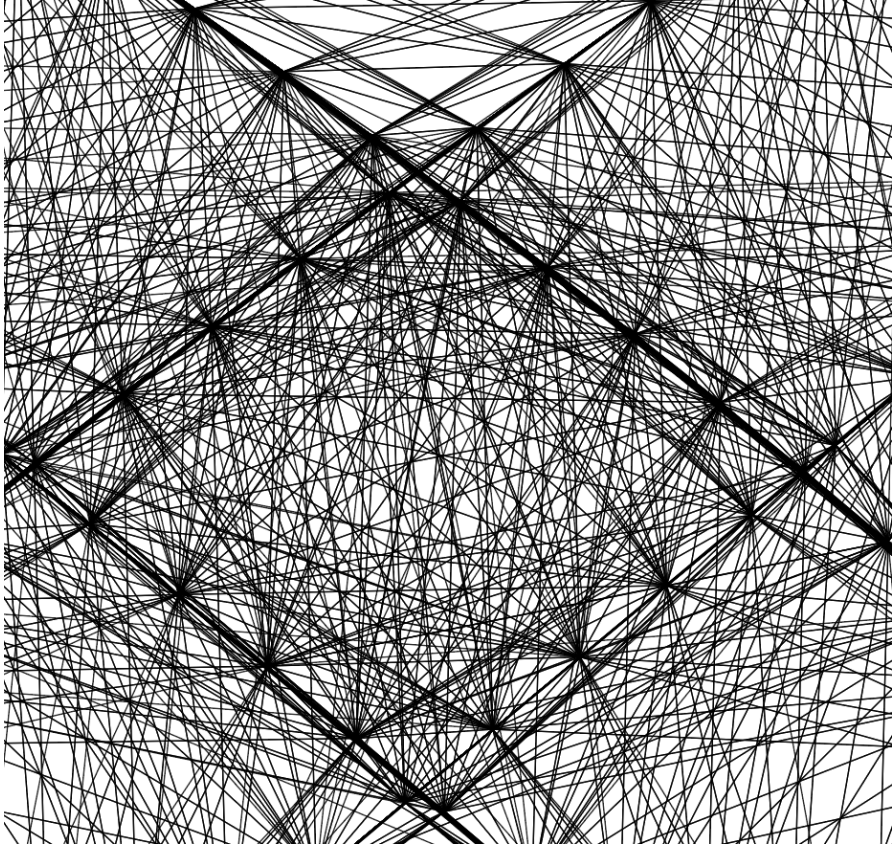


image file

In principle this is the same as vectors in design software where you can keep zooming in



vector

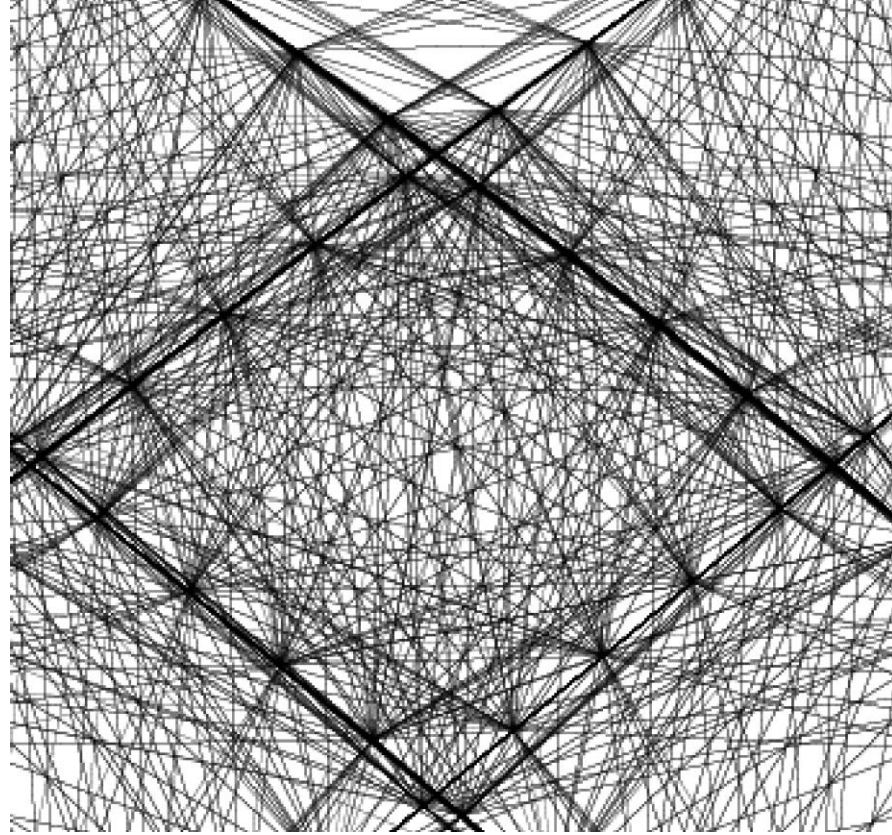
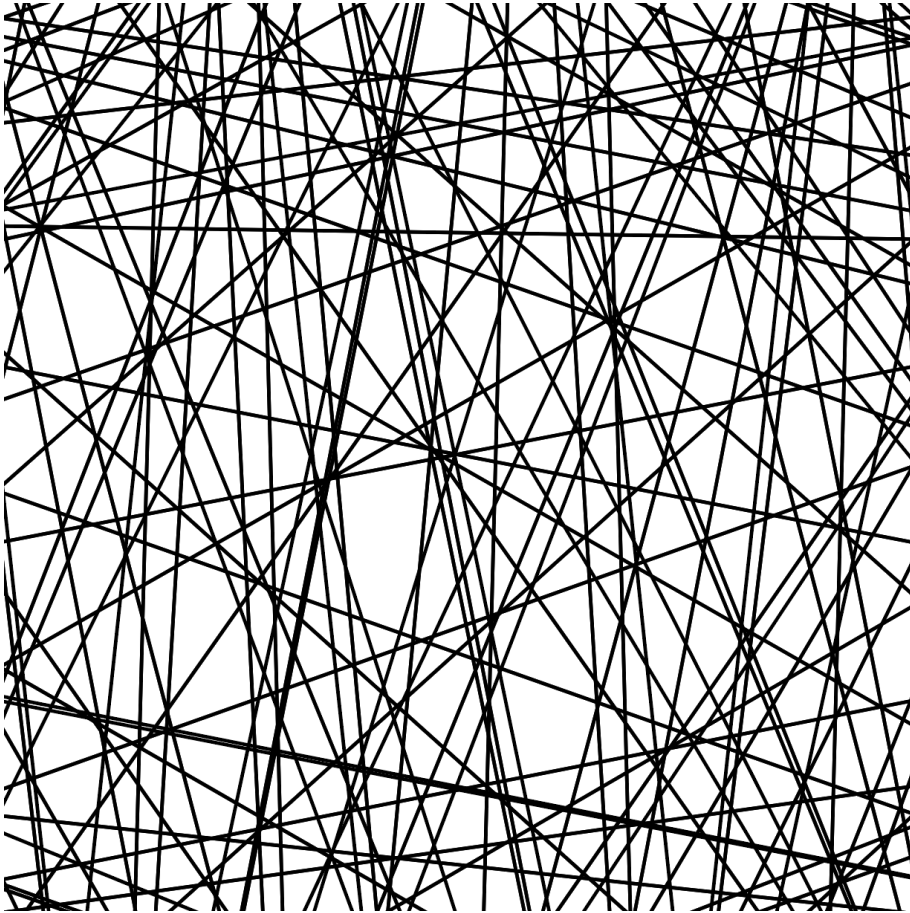


image file

In principle this is the same as vectors in design software where you can keep zooming in



vector

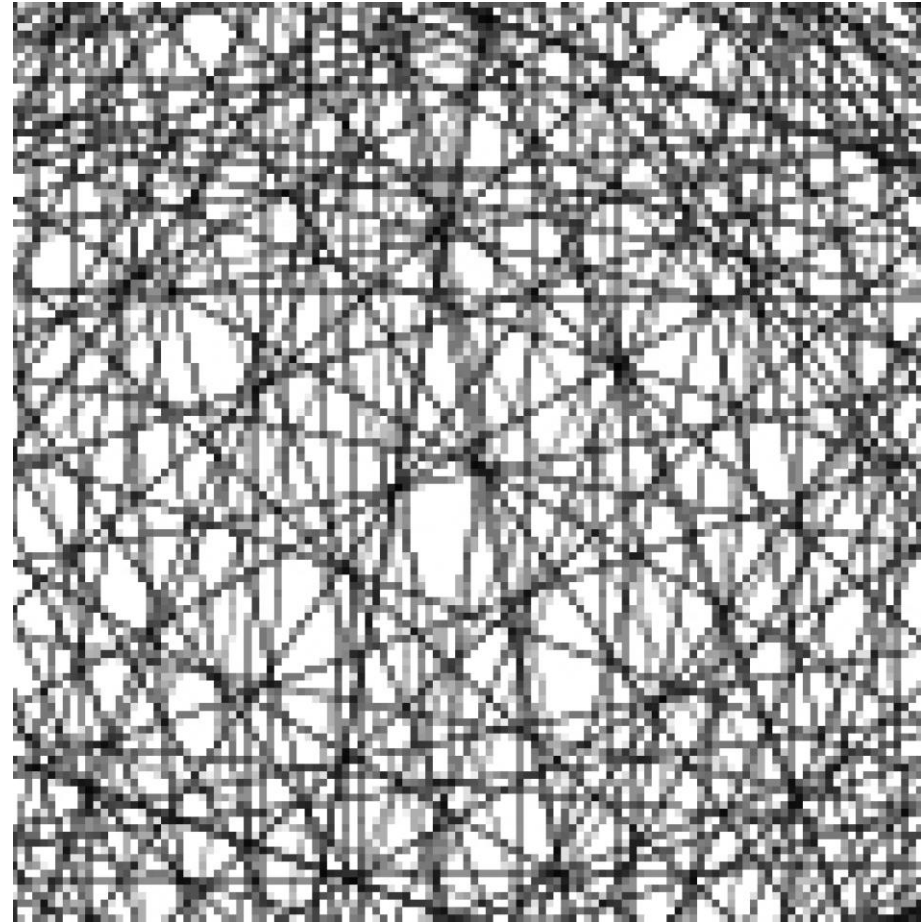
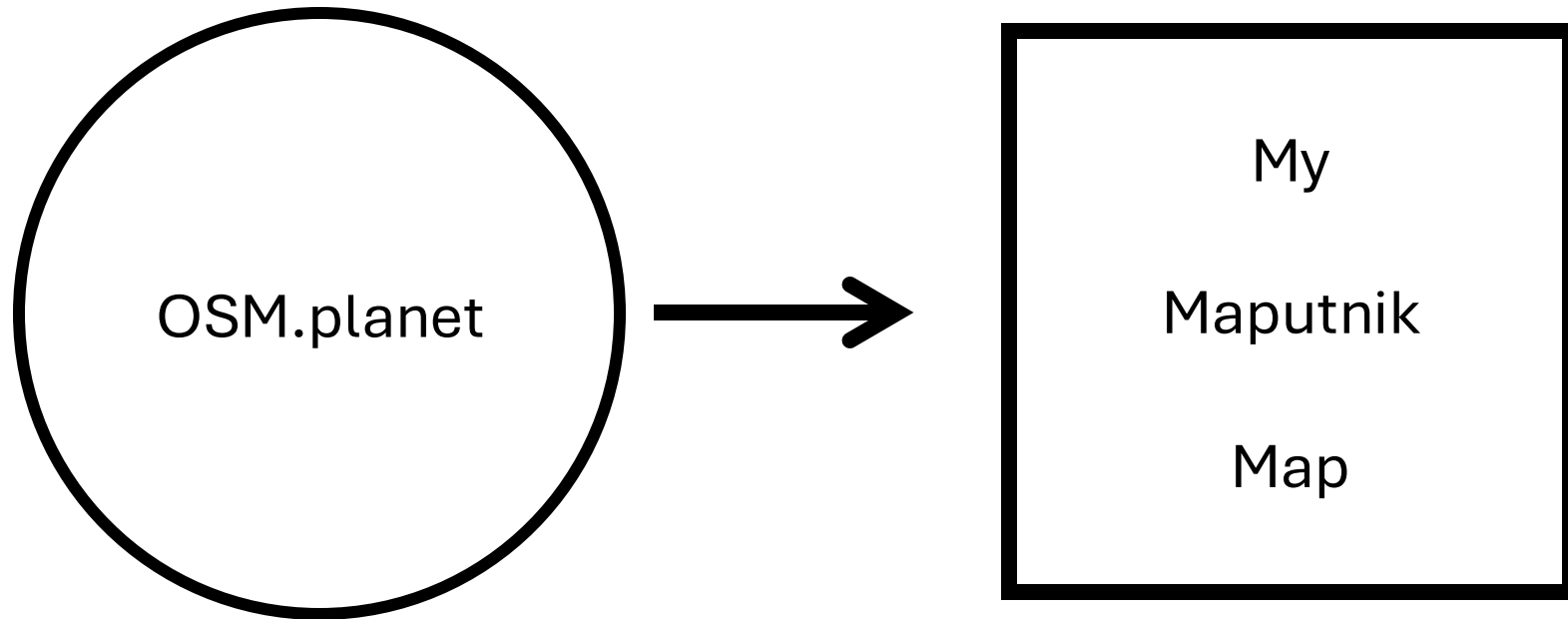


image file

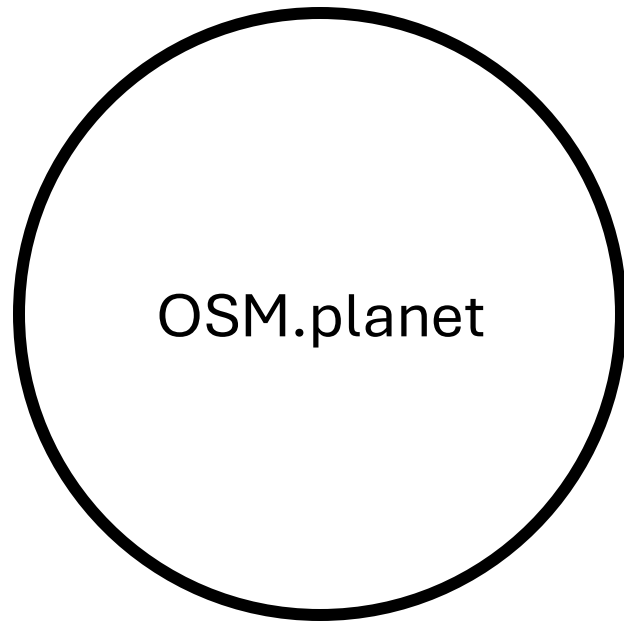
But what is **Maputnik**?

This means it is easier to edit and customize or maps by directly accessing the data we want from OSM and telling the browser how we want it to be displayed



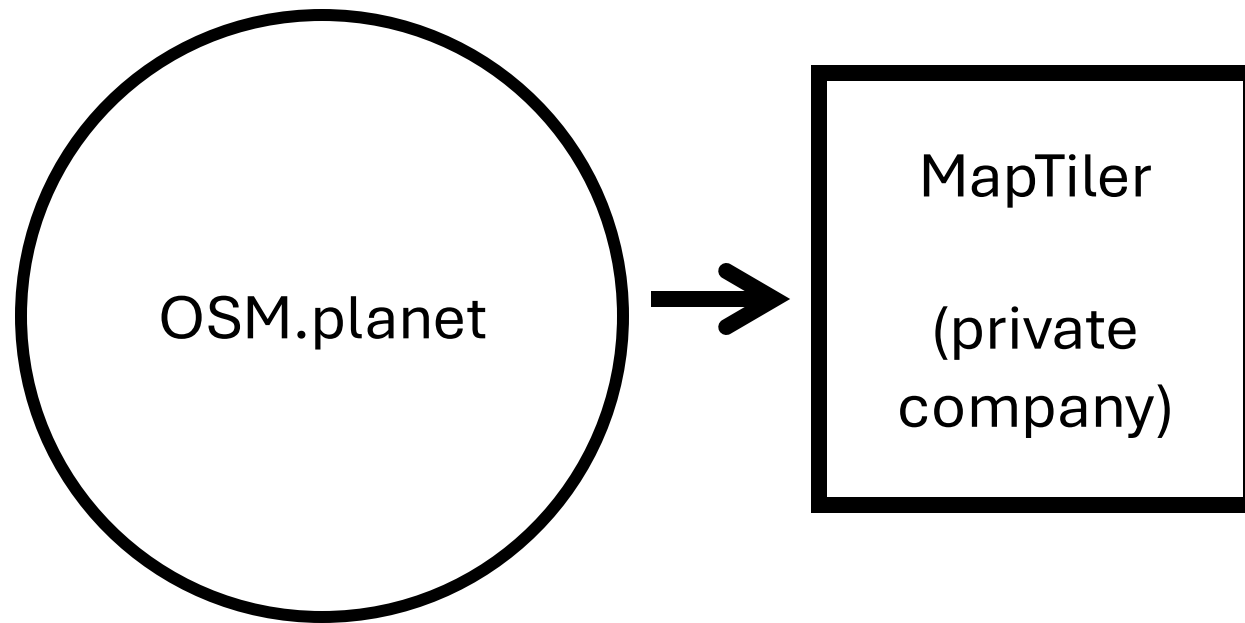
But what is **Maputnik**?

Actually it's a little bit more complicated than that ...



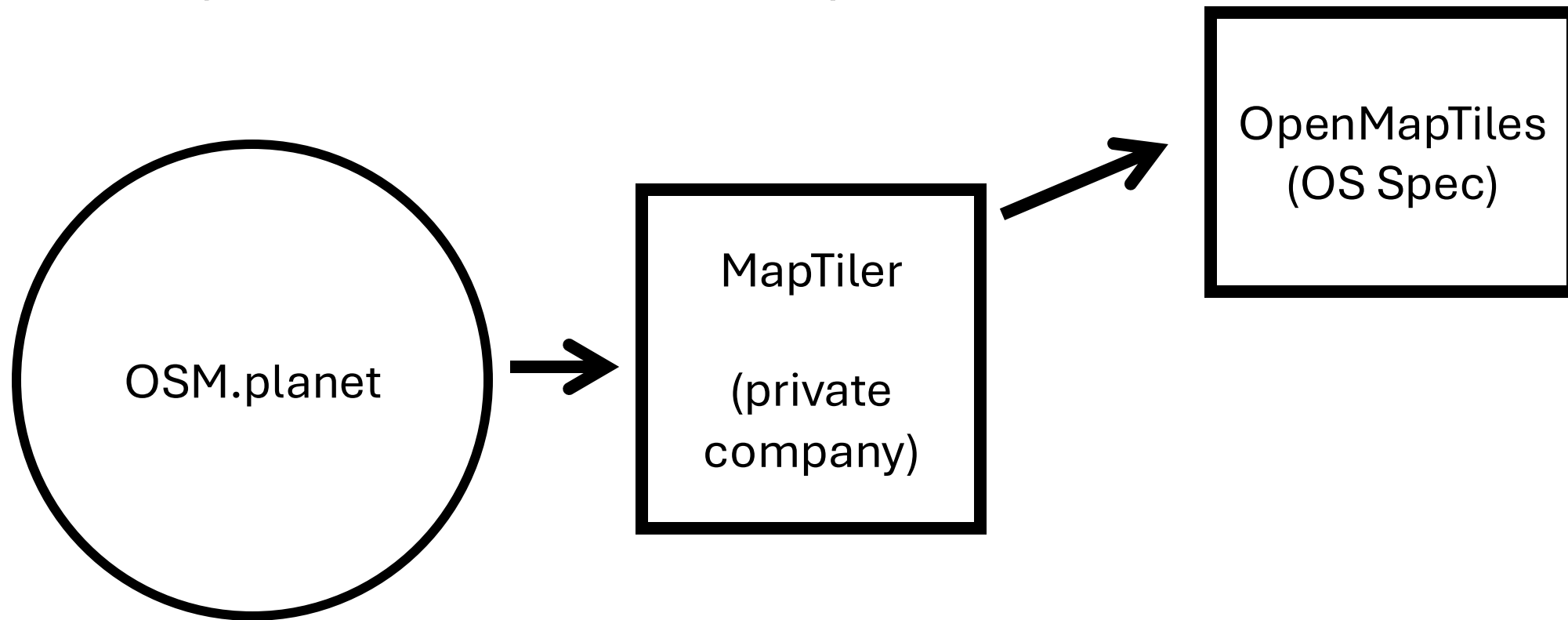
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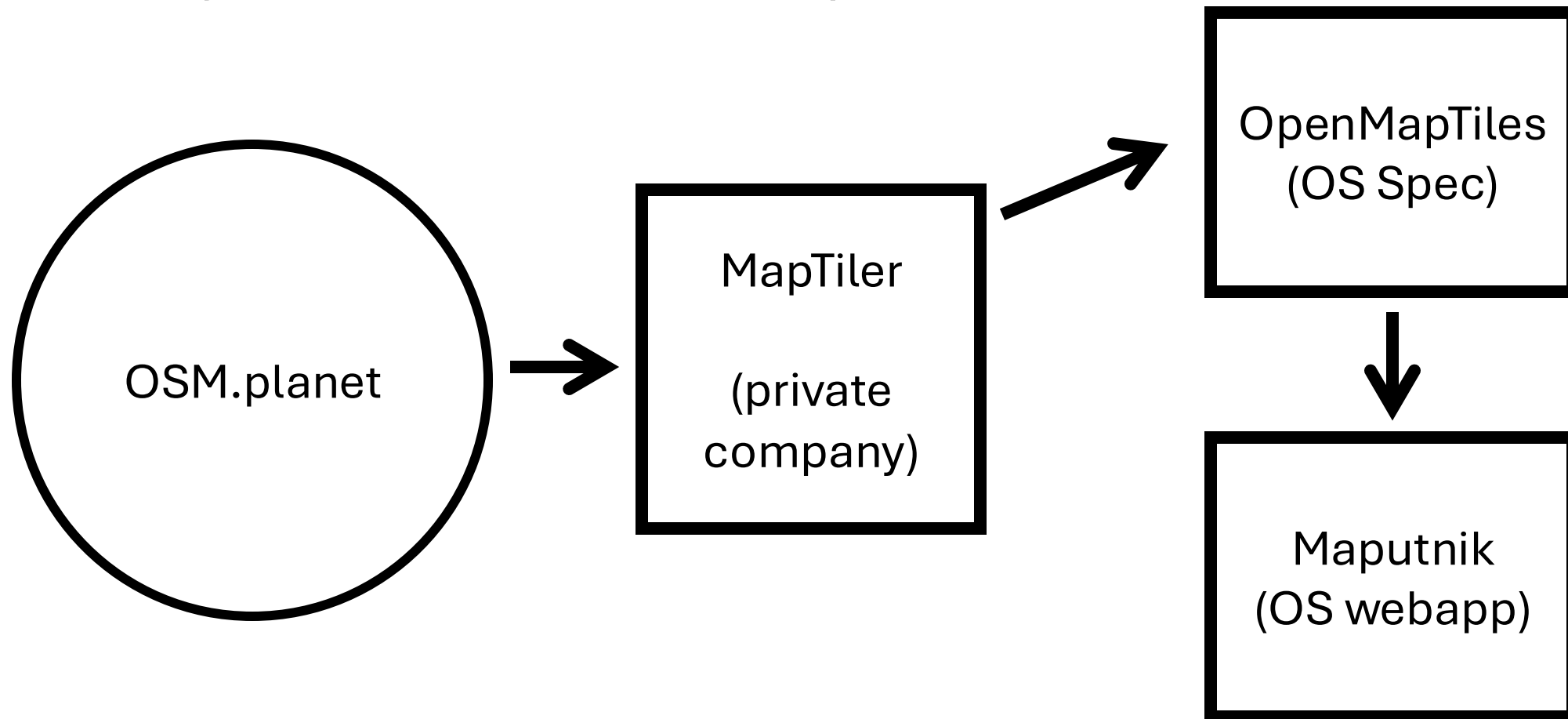
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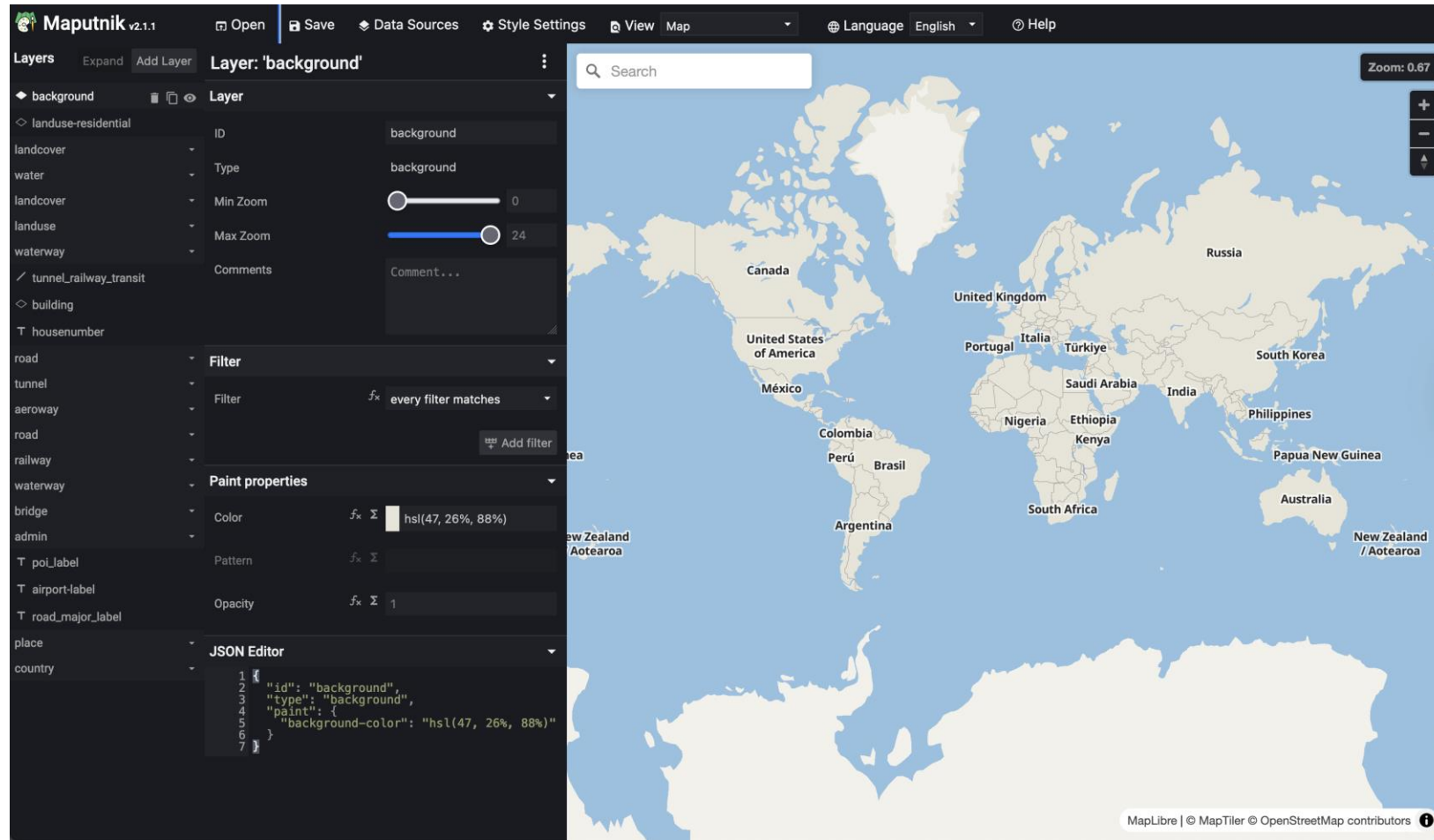


But what is **Maputnik**?

Actually it's a little bit more complicated than that ...

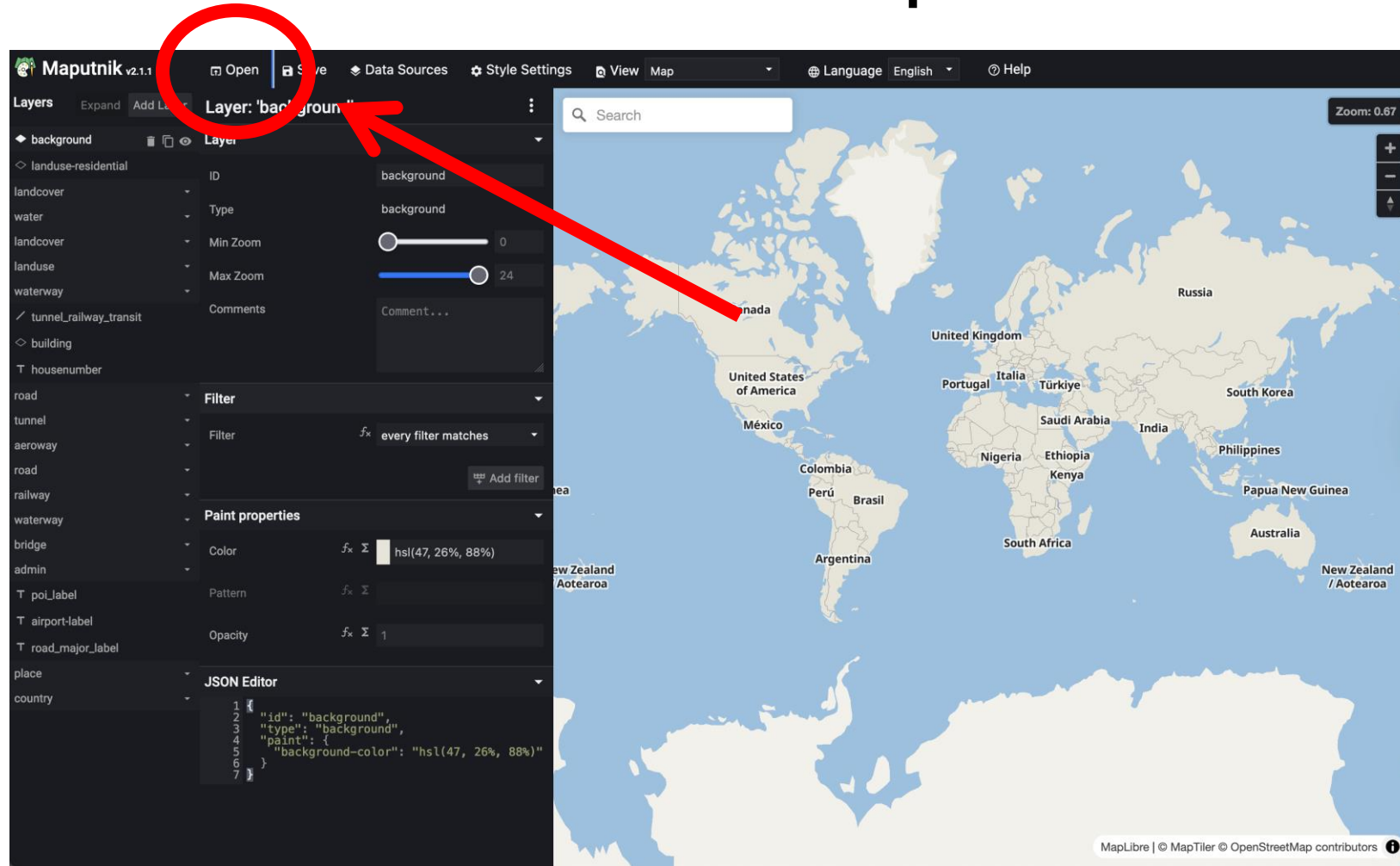


Enough! Lets make a map



<https://maplibre.org/maputnik/>

Lets make a **NEW** map



<https://maplibre.org/maputnik/>

Maputnik v2.1.1

OpenSaveData Source

Open Style

Help

LayersExpandAdd Layer

Layer: 'background'

background

landuse-residential

landcover

water

landcover

landuse

waterway

tunnel_railway_transit

building

housesnumber

road

tunnel

aeroway

road

railway

waterway

bridge

admin

poi_label

airport-label

road_major_label

place

country

Layer

ID

Type

Min Zoom

Max Zoom

Comments

Filter

Filter

Paint properties

Color

Pattern

Opacity

JSON Editor

Open local Style

Open a local JSON style from your computer.

Open Style

Load from URL

Load from a URL. Note that the URL must have **CORS enabled**.

Enter URL...

Load from URL

Gallery Styles

Open one of the publicly available styles to start from.

Empty Style

Americana

Dark Matter

MapTiler Basic

Toner

Zoomstack Light

Zoomstack Night

Zoomstack Outdoor

Zoomstack Road

OSM Bright

OSM Liberty

OSM OpenMapTiles

Zoom: 0.67

+

-

↑

↓

Russia

South Korea

India

Philippines

Papua New Guinea

Australia

New Zealand / Aotearoa

South Africa

Ethiopia

Kenya

Saudi Arabia

Türkiye

Italy

dom

MapLibre | © MapTiler © OpenStreetMap contributors

Lets make a blank map

Maputnik v2.1.1

Open Style

Layers

Layer: 'background'

Landuse-residential

landcover

water

landcover

landuse

waterway

tunnel_railway_transit

building

housesnumber

road

tunnel

aeroway

road

railway

waterway

bridge

admin

poi_label

airport-label

road_major_label

place

country

Open Style

Load from URL

Load from a URL. Note that the URL must have CORS enabled.

Enter URL...

Load from URL

Gallery Styles

Open one of the publicly available styles to start from.

Empty Style

Americana

Dark Matter

MapTiler Basic

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Zoomstack Light

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Zoomstack Road

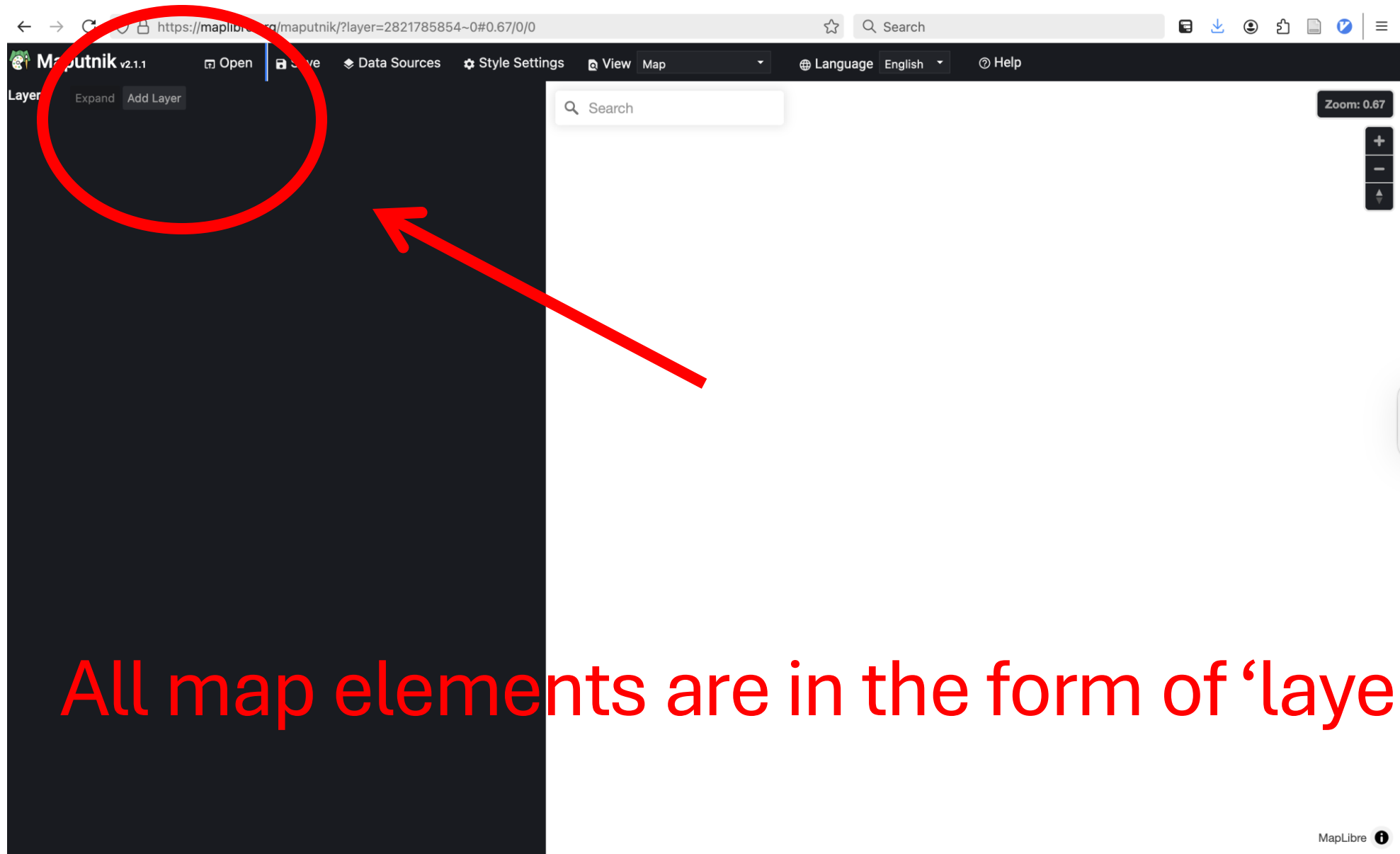
OSM Bright

OSM Liberty

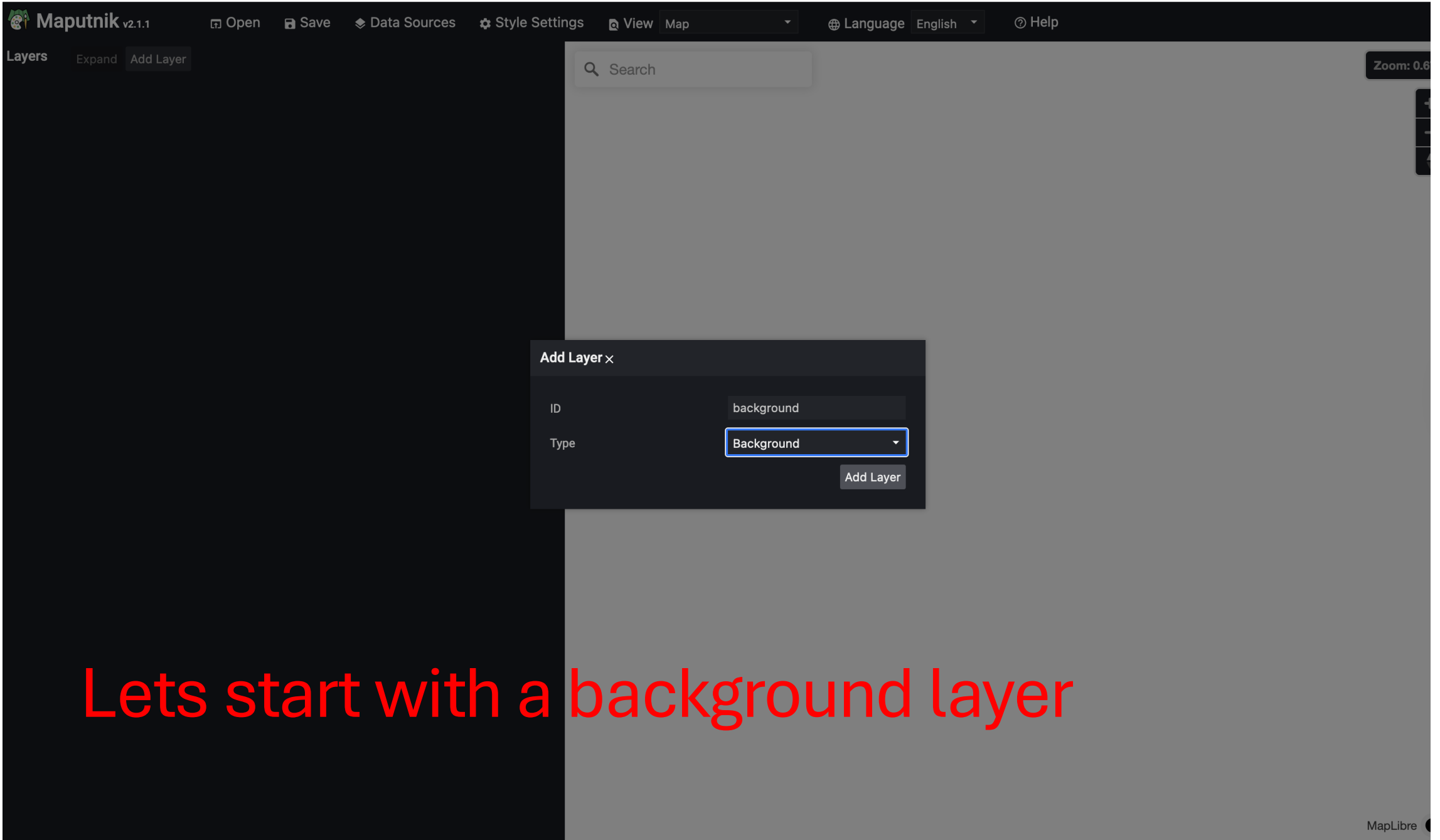
OSM OpenMapTiles

Zoom: 0.67

MapLibre | © MapTiler © OpenStreetMap contributors



All map elements are in the form of 'layers'



Lets start with a background layer

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

Layer: 'background'

background Layer

IDbackground

Typebackground

Min Zoom0

Max Zoom24

Comments

Filter

Filterfxevery filter matches

Add filter

Paint properties

ColorfxΣrgba(44, 166, 213, 1)

PatternfxΣ

OpacityfxΣ1

JSON Editor

```
1 {
2   "id": "background",
3   "type": "background",
4   "paint": {
5     "background-color": "rgba(44, 166, 213, 1)",
6   }
7 }
```

Search

Zoom: 0.67

+

-

↕

Each layer then has lots of properties that we can edit, like color

#2CA6D5

HEX

MapLibre

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

Layer: 'background' ⋮

background

Layer

IDbackground

Typebackground

Min Zoom0

Max Zoom24

CommentsComment...

Filter

Filterfxevery filter matches

Add filter

Paint properties

ColorfxΣrgba(44, 166, 213, 1)

PatternfxΣ

OpacityfxΣ1

JSON Editor

```
1 {
2   "id": "background",
3   "type": "background",
4   "paint": {
5     "background-color": "rgba(44, 166, 213, 1)",
6   }
7 }
```

Search

Zoom: 0.67

+

-

⬆️

Color picker

2C 6.15

HEX

This also includes JSON code of all of this information

MapLibre

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer Layer: 'background'

background Layer

ID background

Type background

Min Zoom

Max Zoom

Comments

Filter

Filter every filter in

Paint properties

Color rgba(44, 166, 213, 1)

Pattern

Opacity 1

JSON Editor

```
1 {
2   "id": "background",
3   "type": "background",
4   "paint": {
5     "background-color": "rgba(44, 166, 213, 1)"
6   }
7 }
```

Source

Active Sources

Choose Public Source

Add one of the publicly available sources to your style.

OpenMapTiles v3 #openmaptiles

Thunderforest Transport v2 #thunderforest_transport

Thunderforest Outdoors v2 #thunderforest_outdoors

OS Open Zoomstack v2 #open_zoomstack

Add New Source

Add a new source to your style. You can only choose the source type and id at creation time!

Source ID kfs9j1f

Source Type Vector (TileJSON URL)

TileJSON URL https://localhost:8080/tiles/monsoon,so

Add Source

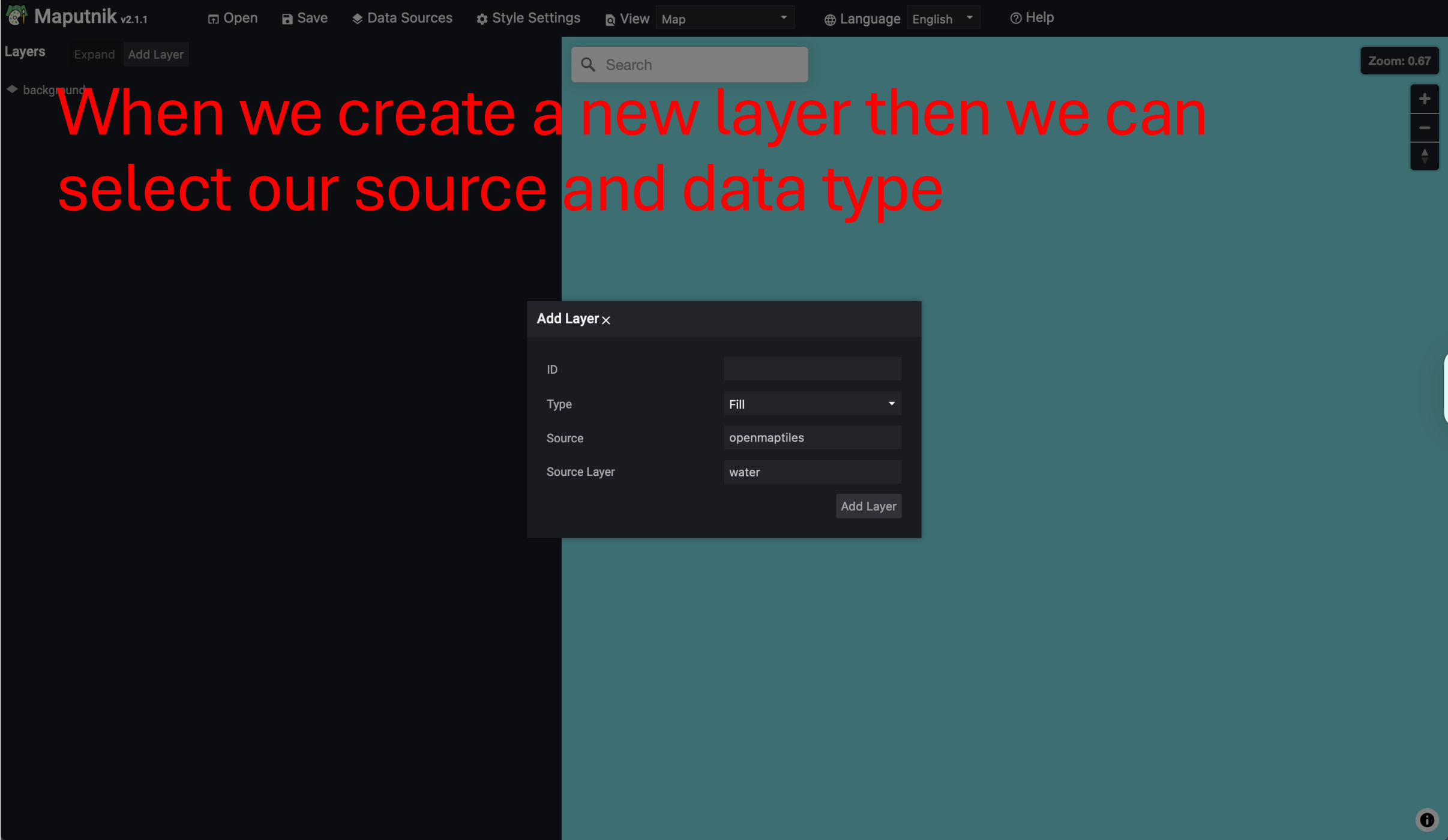
Search

Zoom: 0.67

MapLibre

Lets make a new layer with some data

First we need to add a data source



When we create a new layer then we can select our source and data type

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

Layer: [empty_string]

background

Layer

ID

Type

Source

Source Layer

Min Zoom

Max Zoom

Comments

Filter

Filter

Add filter

Paint properties

Opacity

Color

Antialias

Outline color

Pattern

Translate

Search

Zoom: 0.67

+

-

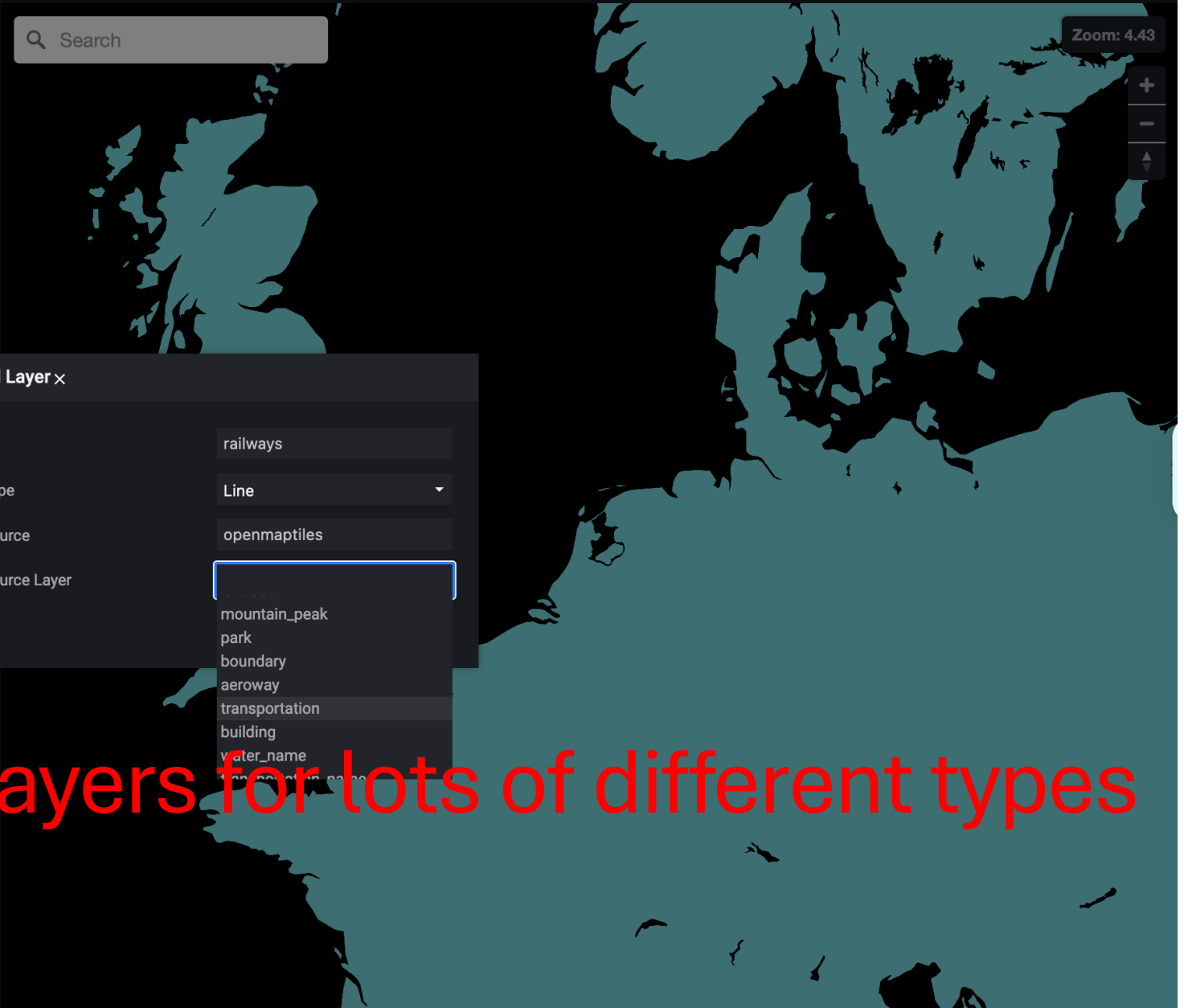
⬆️⬆️



Water is black!

Layers | Expand | Add Layer

- background
- water



Add Layer x

ID	railways
Type	Line
Source	openmaptiles
Source Layer	mountain_peak park boundary aeroway transportation building water_name natural

We can add in layers for lots of different types of OSM data

← → ↺ https://maplibre.org/maputnik/?layer=2423621589~2#4.43/52.69/5.01 ☆ Search

Maputnik v2.1.1 Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer Layer: 'railways'

- background
- water
- / railways

Layer

ID railways

Type line

Source openmaptiles

Source Layer transportation

Min Zoom 0

Max Zoom 24

Comments Comment...

Filter

Filter f_x every filter matches

+ Add filter

Paint properties

Opacity f_x 1

Color f_x #000000

Width f_x 1

Offset f_x 0

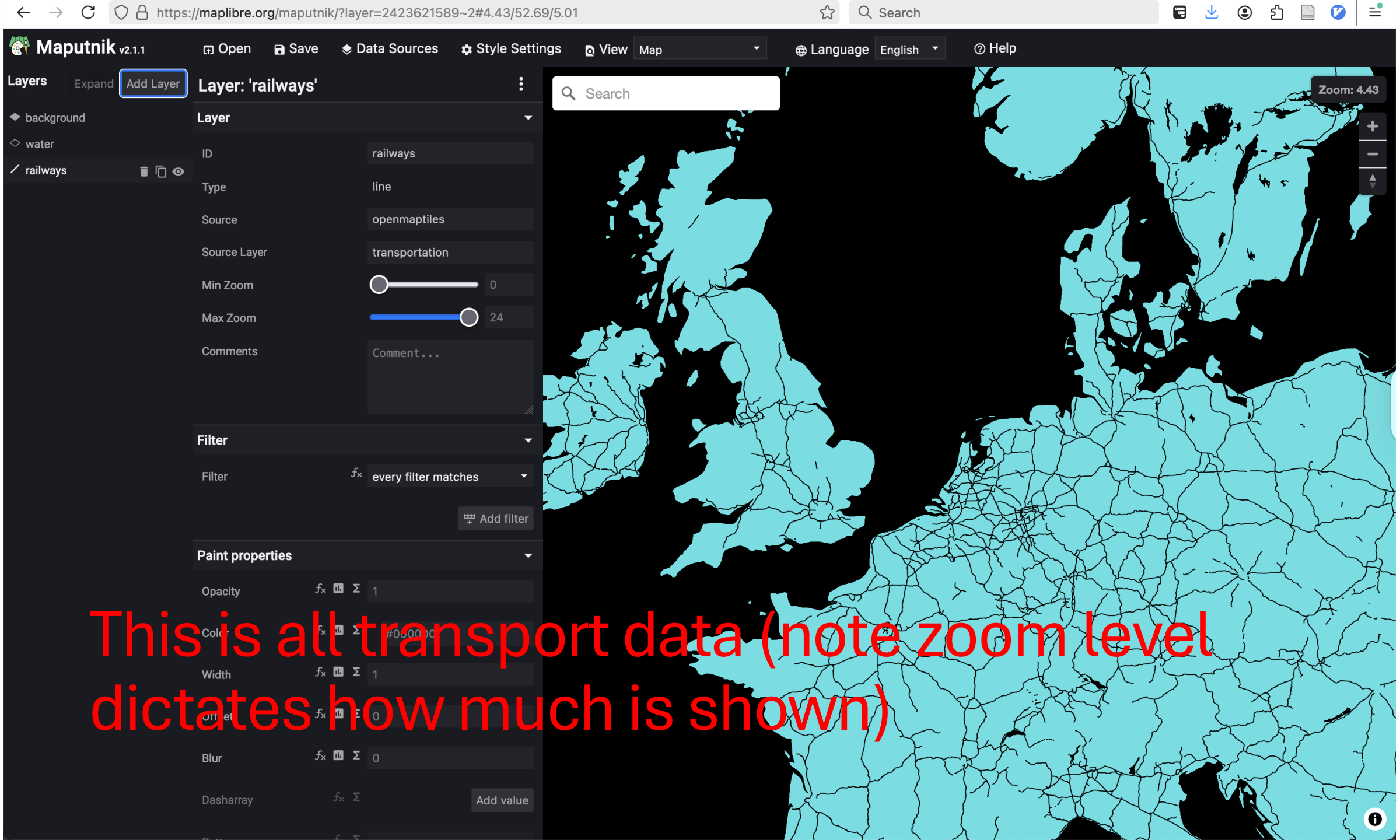
Blur f_x 0

Dasharray f_x Add value

Search

Zoom: 4.43

This is all transport data (note zoom level dictates how much is shown)



Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer Layer: 'railways'

background water railways

Layer

ID railways

Type line

Source openmaptiles

Source Layer transportation

Min Zoom 0

Max Zoom 24

Comments Comment...

Filter

Filter f_x every filter matches

class == rail

Paint properties

Opacity f_x Σ 1

Color f_x Σ #000000

Width f_x Σ 1

Offset f_x Σ 0

Blur f_x Σ 0

Dasharray f_x Σ

Add value

Search

Zoom: 8.08

+

-

↑

↓

We can use filters to specify data within that layer > here we are selecting only railways

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

- background
- water
- railways

Min Zoom 0

Max Zoom 24

Comments Comment...

Filter

Filter every filter matches

class == rail

Paint properties

Opacity f_x Σ 1

Color

Function interpolate

Base 1

Zoom Output value

6	#000000
16	rgba(10, 225, 26, 1)

Add stop f_x Convert to expression

Width f_x Σ 1

Offset f_x Σ 0

Blur f_x Σ 0

Dasharray f_x Σ Add value

Search

Zoom: 8.62

+

-

↑

↓

We can add in further conditions that change how this data is displayed as we zoom in and out

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

- background
- water
- railways

Source Layer: transportation

Min Zoom: 0

Max Zoom: 24

Comments: Comment...

Filter

Filter: every filter matches

class == rail

Add filter

Paint properties

Opacity: 1

Color

Function: interpolate

Base: 1

Zoom Output value

6	#000000
16	rgba(10, 225, 26, 1)

Add stop Convert to expression

Width: 1

Offset: 0

Blur: 0

Dasharray: Add value

Search

Zoom: 12.30

Here our railways turn green when we reach zoom level 16!

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

Background

Residential

Cemetery

Military

Railway

Garage

Dam

Quarry

Industrial

Retail

Commercial

Education and Health

Aeroway

Wetland (medium sca

Sand (medium scale)

Grass (medium scale)

Rock (medium scale)

Wood (medium scale)

Farmland (medium sc

Marsh

Park

Pitch

Stadium

Garden

Wood

Tidalflat

Wetland and swamp

Scree

Min Zoom 7

Max Zoom 24

Filter

Filter f_x every filter matches

class == rail

\$type == Polygon

Add filter

Paint properties

Opacity f_x 1

Color f_x hsl(129, 24%, 73%)

Antialias f_x

Outline color f_x

Pattern f_x

Translate f_x 0

Translate anchor f_x Map Viewport

JSON Editor

```
1 {
2   "id": "Cemetery",
3   "type": "fill",
4   "source": "openmantiles"
```

Search

Zoom: 16.04

+ -

Gibbet Hill Road

Student Cinema

Warwick Business School

Library Road

Cafe Library

Warwick University Library

Centre for Interdisciplinary Methodologies

Cafe Social Sciences

Chaplaincy

Warwick Arts Centre

Warwick Arts Centre Cinema

Warwick Arts Centre Theater

Days of Judgement Cat 1

Academy Way

University Road

Academic Loop

Around Zero

Forest Planet - 3 2009

White Koan

Cafe Oculus

Health Centre Road

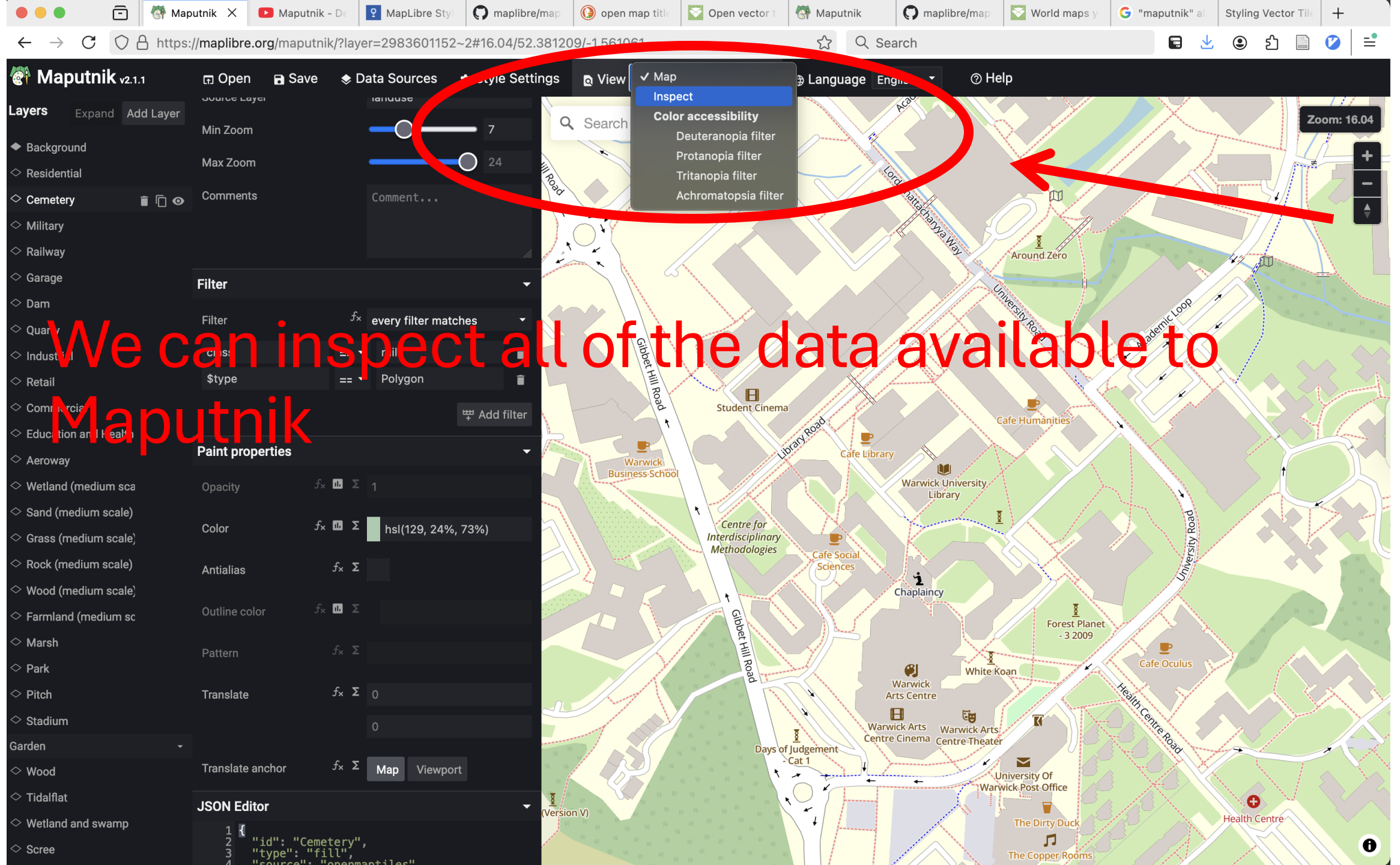
University Of Warwick Post Office

The Dirty Duck

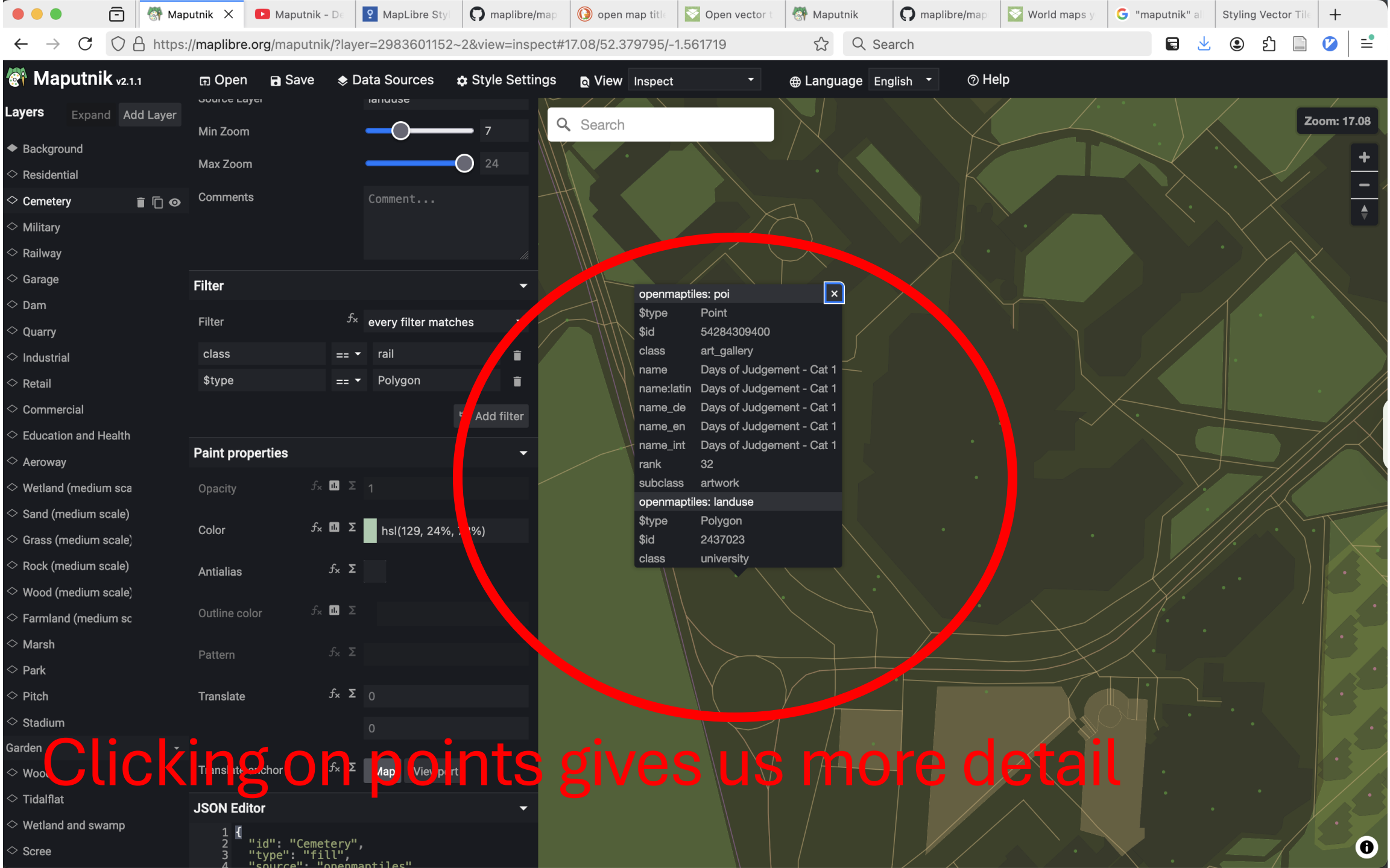
The Copper Rooms

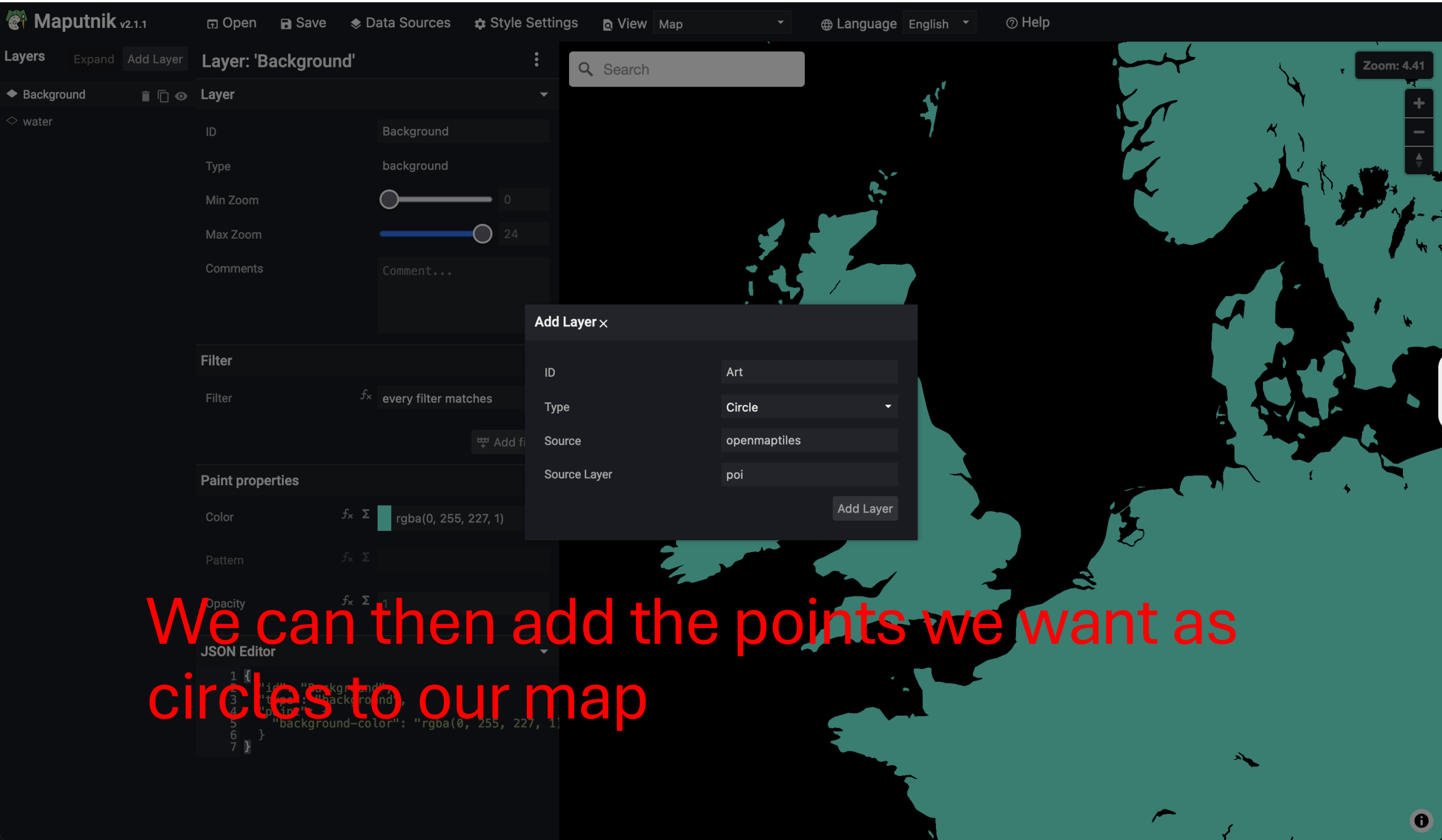
Health Centre

Exploring data sources!

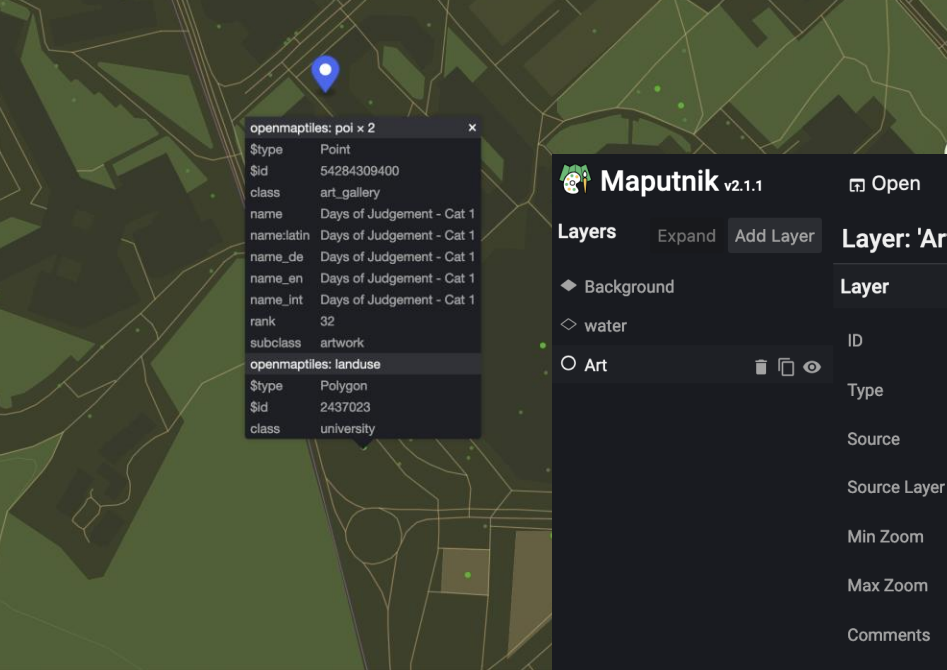


We can inspect all of the data available to Maputnik





We can then add the points we want as circles to our map



openmaptiles: poi x 2	
\$type	Point
\$id	54284309400
class	art_gallery
name	Days of Judgement - Cat 1
name:latin	Days of Judgement - Cat 1
name_de	Days of Judgement - Cat 1
name_en	Days of Judgement - Cat 1
name_int	Days of Judgement - Cat 1
rank	32
subclass	artwork
openmaptiles: landuse	
\$type	Polygon
\$id	2437023
class	university

Maputnik v2.1.1

Open Save Data Sources Style Settings View Map Language English Help

Layers Expand Add Layer

Background

water

Art

Layer: 'Art'

Layer

ID Art

Type circle

Source openmaptiles

Source Layer poi

Min Zoom 0

Max Zoom 24

Comments Comment...

Filter

Filter f_x every filter matches

class == art_gallery

Add filter

Paint properties

Color f_x rgba(255, 255, 255, 1)

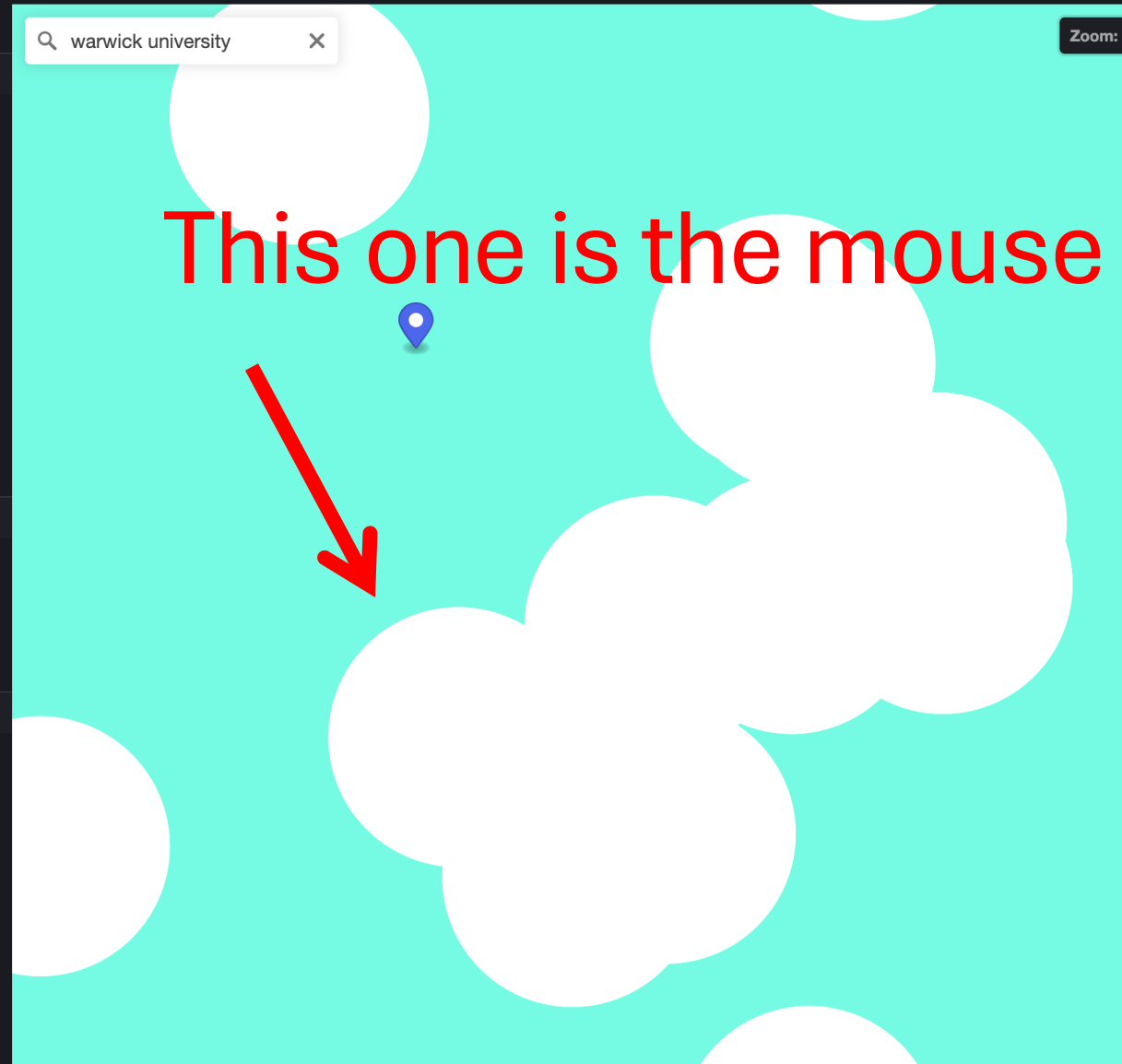
Opacity f_x 1

Stroke color f_x #000000

Stroke opacity f_x 1

Blur f_x 0

Radius f_x 100



This one is the mouse

Thanks, Tim, I can't wait to make exactly the map I want....



Actually Maputnik sort of has a lot of limitations

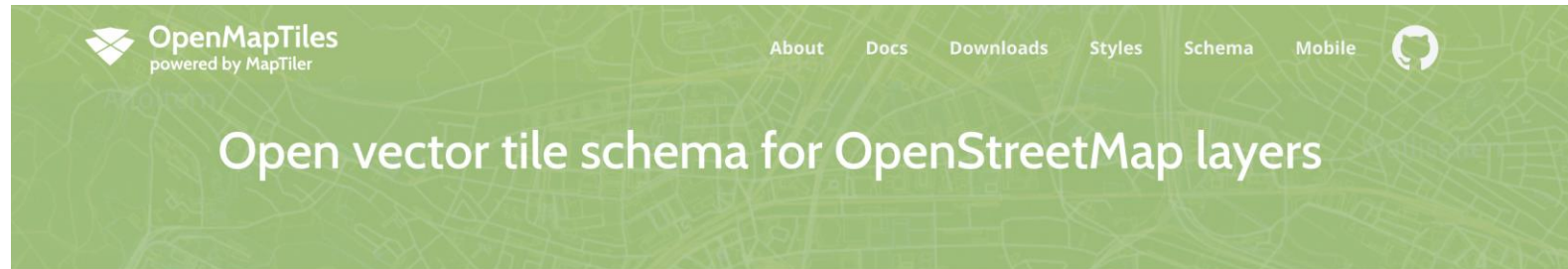
- Limited data from OSM
- Assumptions about crowding and zoom levels
- Fixed iconography

Actually Maputnik sort of has a lot of limitations

- Limited data from OSM
- Assumptions about crowding and zoom levels
- Fixed iconography

Which can all be worked around, if you really want to...

Limited data from OSM



Inspect

Layers

- aerodrome_label
- aeroway
- boundary
- building
- housenumber
- landcover
- landuse
- mountain_peak
- park
- place
- poi
- transportation
- transportation_name
- water
- water_name
- waterway

The vector tile schema describes how the vector data is organized into different thematic layers and which attribute and values each layer contains. This is useful for writing a map style and allows for alternative implementations of a schema.

The OpenMapTiles schema is open (CC-BY) and you are free to use, extend or [build upon the existing implementation](#) (BSD license). Attribution is required. Please properly reference OpenMapTiles if you build upon the schema.

The vector tile schema has been developed by [Klokant Technologies GmbH](#) and was initially modelled after the cartography of the [Carto Basemap Positron](#). The vector tile schema has been refined and improved in cooperation with the [Wikimedia Foundation](#) and is heavily influenced by the many years Paul Norman's experience of creating maps from OpenStreetMap.

Schema [updates](#) and [changelog](#).

Definition of layers

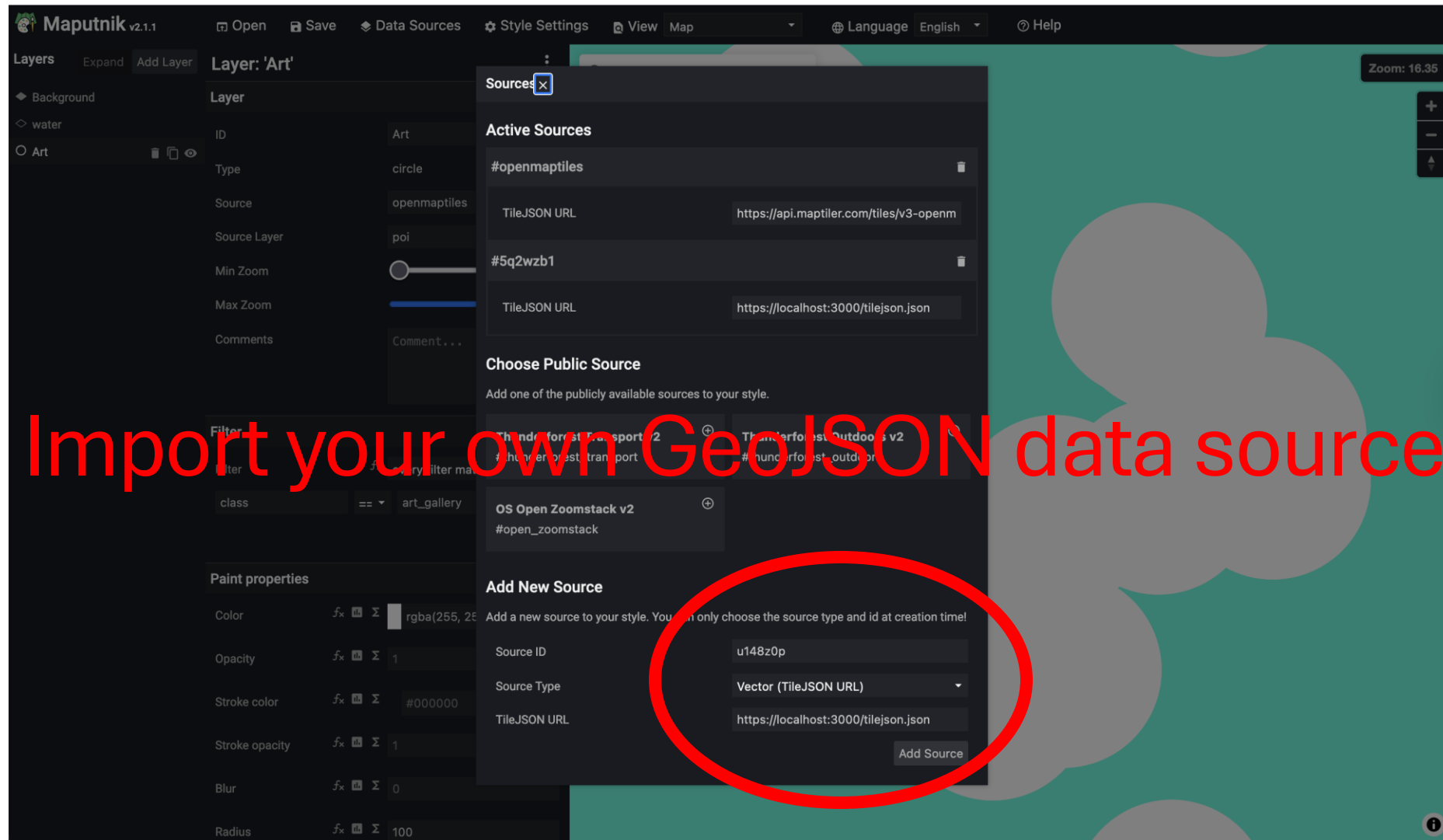
- aerodrome_label ↗

[Aerodrome labels](#)

Show Diagrams

View on GitHub

Work around



Work around

- Find the feature you are looking for (overpass / taginfo)
- Query and download the data from overpass
- Host the GeoJSON somewhere yourself (could be github)
- Import it as a layer in Maputnik

Zoom levels

- Convert the geographical data you want to GeoJSON
- Use [Tippecanoe](#) (a tool for converting geodata to vector files)
command line to adjust the underlying zoom level assumptions

```
tippecanoe -o output-file -Z zoom_level_in -z  
zoom_level_out input_file
```

- Host the .mbtiles output on maptiler or self-host as a data source

(see: <https://www.theinformationlab.co.uk/community/blog/zoom-levels-mapbox/>)

Iconography

- Icons are not just an image
- Icons are a mix of one image and a separate JSON file – which Maputnik then looks up

A Sprite typically consists of 4 files:

- `sprite.png` - the image with all the icons, for example:



- `sprite.json` - file describing the names and positions of the icons in the image (the names are later important when modifying the style in the Customize)
- `sprite@2x.png` and `sprite@2x.json` - the same as above, but in higher resolution (HiDPI/retina)

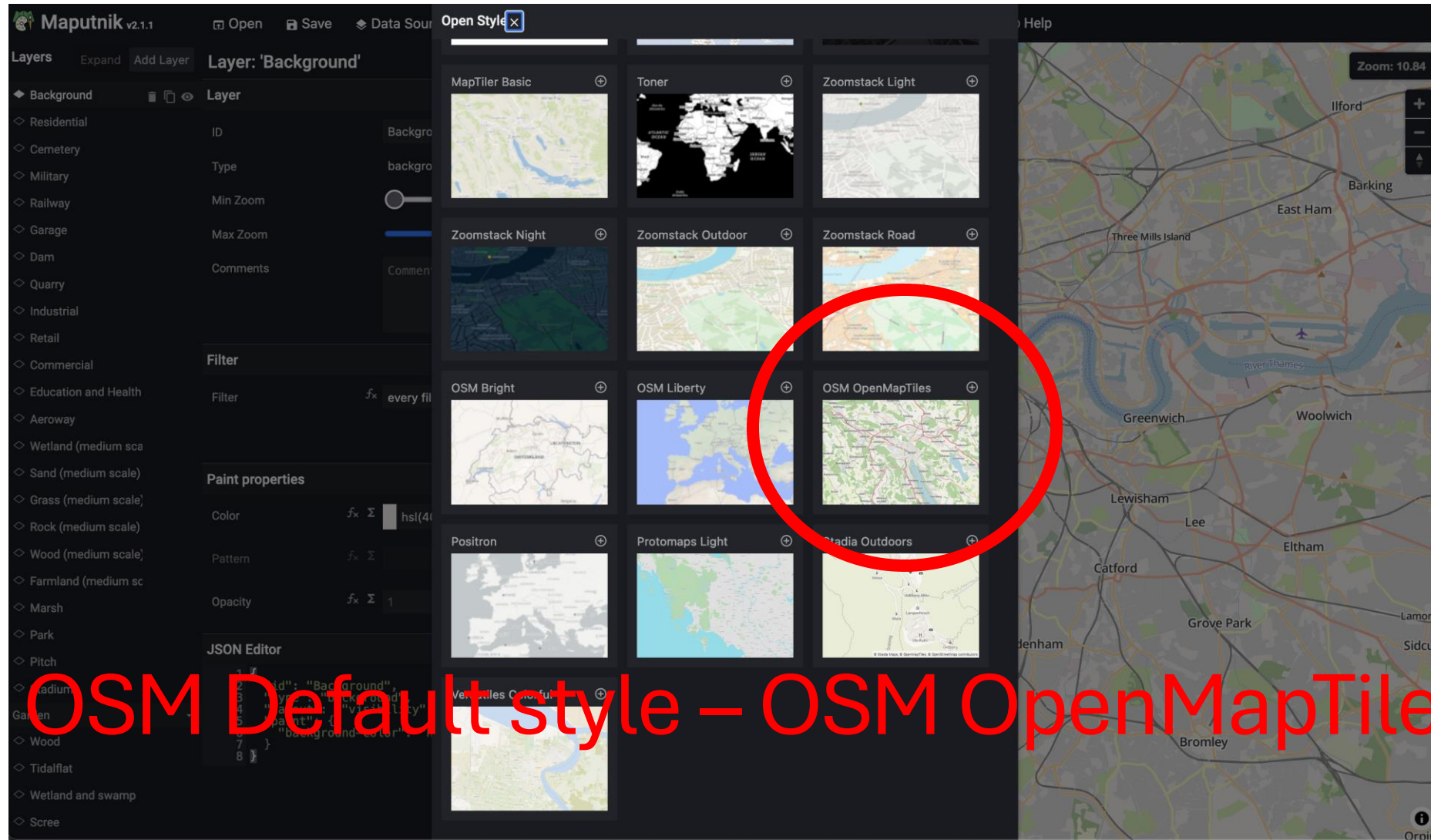
Iconography

- You can create your own sprite packs using command line tools like [Spreet](#)
- You need to do this for **ALL** the symbols on your map at once
- Then you need to host these files (as an API!) for Maputnik to query

OpenSource software as a service

- While MapTiler is the driving force behind Maputnik and lots of these software – and this is to be applauded – they also sell services that use this software
- Notably, lots of these ‘problems’ are more easily fixable if you pay for MapTiler services or consult with them to make a map for you
- ...

What we are going to do today



What we are going to do today

- Try editing the default map to make it closer to what we want
 - Delete unwanted layers
 - Try adding in new ones
 - Changing colors and styles of existing layers
- Making a (post-it) note every time we encounter
 - Can't change something we want to
 - Find something we want to add (but isn't there on Maputnik)
 - Different symbols we would want to use (sketch them out!)
- Save your resulting JSON file!